

Modal definability in precompact polyhedra

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The aim of the paper is to prove an equivalent of Goldblatt-Thomason's theorem for classes of precompact polyhedra, i.e. subsets of the affine space whose closure is a compact polyhedron.

We adapt from the semantics from [1] on compact polyhedra, where the closure operator $\Diamond$ acts as a closure on subpolyhedra, to draw a closer connection between polyhedra and what we call 'simplicial precomplexes', some combinatorial objects, that generalize simplicial complexes.

This allows us to swiftly translate conditions between polyhedra -as geometric objects, precomplexes - as combinatorial objects, and posets, giving us a result similar to [2]

1. Any class of polyhedra is equivalent to the realization of a class of simplicial precomplexes closed under subdivision.
2. A class of (finite) simplicial precomplexes is modally definable if and only if it's closed under disjoint unions, open simplicial images, and open subcomplexes
3. A class of polyhedra is modally definable if and only if it's closed under disjoint unions, open PL-maps and open subpolyhedra.

As simplicial complexes (and their realization) carry topological meaning, we can obtain a negative result on homological properties:

4. The homology chains of a modally definable class of simplicial complexes must be closed under direct sums and images

Thus excluding the definability of any class of the kind "polyhedra of a certain genus"

References

- [1] Nick Bezhanishvili et al. Geometric Model Checking of Continuous Space. In: CoRR abs/2105.06194 (2021). arXiv: 2105.06194. url: <https://arxiv.org/abs/2105.06194>.

- [2]